

"Begin with the End in Mind": Incorporating UX Evaluation Metrics into Design Materials of Participatory Design

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ABSTRACT

Participatory Design (PD) aims to empower users by involving them in various design decisions. However, it was found that the PD's evaluation criteria are usually set by the product team and used only at the end of a design process, without adequate user participation. To address this issue, we proposed introducing UX evaluation metrics into design materials at the participatory design INPUT phase. Using a case study of designing a chatbot for community members to report safety incidents, we studied the impact of this approach with 58 participants from two workshops. Our results showed that the integration of UX evaluation metrics efficiently rationalized participants' contributions and helped identify key evaluation metrics when setting values for new AI systems, enhancing PD workshop insights. In addition to examining the use of the *Program Theory Model* to explain PD, our empirical investigation added a new dimension to this model.

CCS CONCEPTS

• Human-centered computing → Participatory design.

KEYWORDS

Participatory design, UX, evaluation metrics, program theory, design material

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1 INTRODUCTION

The AI and HCI groups are increasingly advocating for the participation of stakeholders and end-users in designing, developing, and maintaining AI systems [8, 28, 51], with the aim of creating AI systems to reflect better the values, preferences, and needs of its users [19, 24]. Participatory design (PD) is a democratic approach for designing social and technological systems that incorporate human activity, based on the idea that users should be involved

in designs they will use and that all stakeholders, especially users, should have equal input into the interface design [38]. In recent years, researchers have frequently adopted this design technique for designing AI-based systems [8].

Participatory design is an approach that focuses on processes and procedures of design [45]. When viewed through the lens of the *Process Theory Model* [19], PD can be broken down into three distinct parts: *input* (the resources required to initiate and complete the PD), *process* (design principles and design activities), and *effects* (the final outputs and its results). As PD informs how processes are used and how effects are thought of, one of its central concerns is how can we evaluate the results of a design *process* in relation to its final *effects* [19]?

Although many participatory design studies have claimed that the application of PD can democratize their design process, a review of PD evaluations found that users are often excluded from the process of defining evaluation criteria for the system [4]. Even in most recent literature, participants are still only consulted as sources when assessing the effects of the system at the end of the PD process, rather than being actively involved in setting these evaluation criteria [37, 41]. This lack of agency in deciding evaluation criteria undermines the democratizing goals of PD [4, 19]. This problem **motivates** us to explore the connections between PD *process* and *effects* that participants are expected to experience. A deeper understanding of the relationships can help PD offer more contributions to the design.

In this paper, we propose incorporating UX *evaluation metrics* (e.g., [53]) into the design materials of participatory design. Through a case study of designing a chatbot for community members to report safety incidents, we examined this new approach in two workshops with 58 participants. Participants were involved in making design decisions and determining the evaluation criteria for the final chatbot.

Our contributions are multifaceted. First, we proved that our case study's participants' preferred design alternatives are correlated with evaluation metrics; hence, incorporating evaluation metrics into the design materials during the *input* phase can benefit in rationalizing the design *effects* generated during the design *process* as shown in Fig.1. Second, empowering participants to define UX evaluation metrics that align with their values by considering the end goal at the start of the design process can provide a more direct and personalized evaluation approach, which may ultimately lead to more successful design outcomes from PD. These decisions, when combined with the established findings from the existing literature, may also help to identify distinct user expectations in novel design contexts. Lastly, to the best of our knowledge, we

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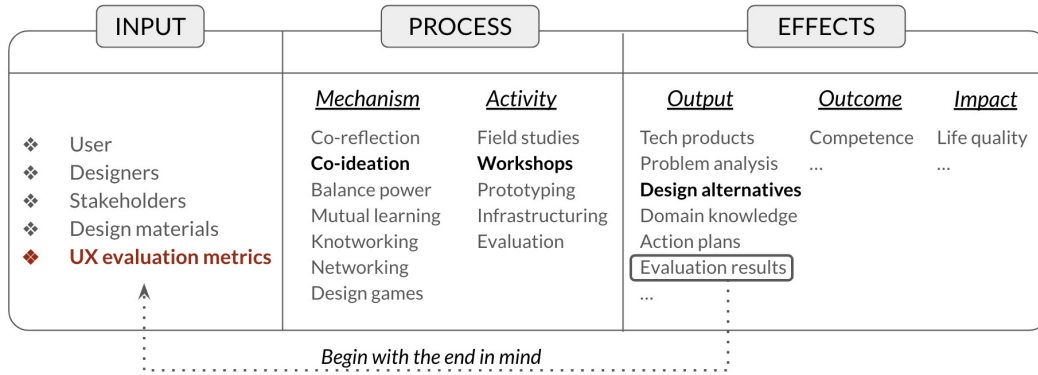


Figure 1: We propose to expand the *Program Theory Model* [19] by adding a new Participatory Design INPUT, i.e., the *UX evaluation metrics* that will be later used to examine the EFFECTS of the design.

are the first to present an empirical example of the introduced usage of the *Program Theory Model* to explain PD [19], and our research added a new dimension to this mode by incorporating UX evaluation metrics into the PD *input* phase.

2 EMPOWERING PARTICIPATORY DESIGN

Participatory design (PD) is an approach to designing technology that involves end-users and stakeholders in the design process, aiming to address the gap between the designers’ understanding of users’ needs and the users’ actual needs by involving them in the design process [38]. PD has risen in popularity in HCI research over the last two decades. The number of PD papers in the ACM digital library has increased by 580% (2003-2013) [35], and 68% of CHI articles have used the word “participatory design” in the last decade [19]. Valuable research questions have been addressed and developed for the PD concerning the issues posed by AI [5, 42], the discussion of meaningful outcomes such as mutual learning [10], the evaluation strategies [4, 30], and a few frameworks to describe the PD’s mechanisms and tactics [8, 18, 19, 30].

Recently, Hansen et al. [19] proposed to construct PD diagrammatically by applying *Program Theory Model*, as shown in Fig. 1. The original model aimed at explaining how programs or project are assumed to work through detailing the causal relations between inputs, processes and effects in those programs [43]. This gives researchers a strong theoretical basis for thinking critically about how, if, and why their activities lead to empowering effects.

In a literature analysis conducted over 24 years by Bossen et al. [4], researchers discovered that in 66 PD investigations, participants had minimal involvement in the crucial stages of defining evaluation criteria. Participants are merely consulted during evaluations through interviews or as information sources, as opposed to being given the power to set evaluation criteria [4, 19]. This contradicts the purported democratic goals of PD and shows that participants are not fully empowered. Also, formal evaluations of PD are typically led by researchers rather than the participants themselves [4], and these evaluations are still rare and generally lack details on methods [16, 35].

Involving participants in the process of defining evaluation criteria may help in ensuring that the chosen metrics are relevant

and meaningful to them. This may also boost the design process’s accountability, leading to greater buy-in and ownership of the final product. In this paper, we apply the diagrammatic illustration of *Program Theory Model* for PD [19], in an effort to contribute to the PD *input* phase, using a reverse engineering logic— by putting one of the components from *effects* to the *input* phase of PD. *How can adding UX evaluation metrics into the design materials enhance understanding to the participatory design effect?* We aim to contribute to the advancement of empirical practices in this field.

3 METHOD

In the following section, we will describe the structure of the participatory design workshops, the design material, and the method of data analysis.

3.1 The Workshops

In Spring 2022 and Fall 2022, we conducted two participatory design workshops with undergraduate students (freshmen and sophomores) who took the HCI courses at our university. There are 58 participants, 26 undergraduates from Spring 2022/Workshop A, and 32 undergraduates from Fall 2022/Workshop B. None of our participants had previously been exposed to chatbot design.

The workshops aimed to improve a chatbot’s designed features for campus safety reporting. Before conducting workshops, a basic chatbot with minimal design feature was implemented and deployed as a conversational-based safety reporting system. Examples of safety concerns were given, such as hate speech, harassment, violence, mental health issues, etc. The chatbot can help collect information from community members and help report the incidents to public service providers accordingly, e.g., police, social workers, or the counseling service on campus.

During the workshops, participants were initially introduced to the objectives and process of the participatory design activity. The primary PD activity was completing a survey, in which participants were provided background about the design context, asked for input on the ideation of design choices and evaluation metrics, and then asked to reflect on further concerns that needed to be addressed. The main difference between the two workshops was that Fall

2022 participants had the opportunity to test the chatbot before completing the PD (Fig. 2).

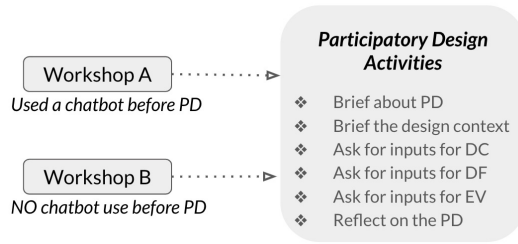


Figure 2: The flowchart for the workshops.

3.2 Design Materials

The design input was presented in plain language to ensure comprehension by all participants, regardless of their technological understanding. We creatively included UX evaluation metrics as a novel design input. Participants rated each input based on perceived importance on a scale of 1 to 5, with 5 being extremely important and 1 being not at all important. They were also prompted to justify their choices through free-form input (e.g., *why do you make such a choice?*). The following section introduces the three major design inputs: design characteristics as the overarching traits of a chatbot, design features as the specific functionalities and capabilities of the chatbot, and evaluation metrics as the measures used to assess the effectiveness and success of a system or product.

- **Design characteristics (DC).** This section includes eight questions about chatbot design characteristics adapted from chatbot surveys [15, 39]. It includes multiple choices for chatbot's tone, empathy, user interaction, integration with service providers, human representation, integration platform, and voice configuration, for example, *"Do you think the chatbot should be able to introduce a human representative into the chat, and why?"*. It also includes one interval scale question for intended roles, i.e., *"Based on a scale of 1-5, how important do you think of the following five roles that you want the chatbot to play in the safety reporting context?"*
- **Design features (DF).** We asked the participants to rate the importance of 12 design features derived from prior research on chatbots relevant to the current safety context. These design features focus on providing adaptive [27, 46], facilitating Q&A [25], using emojis [13, 52], steering conversation topic and flow [22], adding human supporter [31], creating an empathetic profile [33], conducting social dialogues such as self-disclosure and social explanations [12, 47, 50] and small talks [32, 44], having emotional intelligence [2, 40, 48], aligning to individual's chat style [20, 49], and having a social identity [33, 47]. For example, *Tell us how important it is for the chatbot to demonstrate emotional intelligence under this context, and why?*
- **Evaluation metrics (EV).** We surveyed participants to gauge their perceived importance of 20 metrics for evaluating both user behavioral data captured from the chat log (e.g., time

spent on reporting different safety concerns, chat length, Q/A quality, user entry time, error rate, dropout rate, etc.) and *user perceptions data* gathered from the preference survey (e.g., perceived communication effectiveness, perceived appropriateness of agent's social behavior, perceived emotional competence, perceived satisfaction, privacy concerns, information quality, perceived ease of use, usefulness, etc), derived from the chatbot UX survey by Zheng et al. [53].

3.3 Data Analysis

We analyzed the data by first reviewing all qualitative inputs for an overall understanding of participants' experiences. Then, statistical analysis was conducted on the quantitative inputs from the survey. The data was initially analyzed using the Kolmogorov-Smirnov test to check for normality [11], revealing that the sample did not meet this assumption. Thus, non-parametric methods were employed as these methods do not rely on normality assumptions.

To investigate whether design characteristics (DC) and design features (DF) can lead to significantly different ratings for the evaluation metrics (EV), we ran Spearman Correlation Coefficient [7] to determine the strength and direction of the relationship between the variables.

To generate more insights, we further classified participants into groups based on their selection for certain design characteristics (DC) and compare groups differences on design features (DF) and evaluation metrics (EV). Specifically, we classified participants into two groups (YES/NO) based on their selection for the DC: *"Would you like the chatbot to show empathy?"* We ran Wilcoxon Mann-Whitney test [14] for comparing two group means following by the effect size analysis using the Cliff's delta [29, 34], which is widely used to report the effect size of Mann-Whitney U test [29, 34]. Cliff's delta runs from -1 to 1. In general, a Cliff's delta of 0.147 is a small effect, 0.33 is a medium effect, and 0.474 or above is a large effect.

To investigate the two group differences (Workshop A and Workshop B), the chi-square test was employed to compare the differences in design characteristics (DC) between the two groups, while non-parametric Mann-Whitney tests [14] were used to evaluate design features (DF) and evaluation metrics (EV) between the two groups. Once we observed significant differences, the Cliff's delta were calculated for the effect size.

4 RESULTS

In this section, we will present the survey questionnaire's statistical analysis results.

4.1 Using Correlation of Evaluation Metrics and Design Alternatives to Prioritize Features

With the use of the Spearman Correlation Coefficient, we investigated the existence and strength of potential relationships between 1) the numerical variable in design characteristics (i.e., DC2 – *On a scale from 1 to 5, please indicate the perceived relevance of the chatbot's various roles: information collector, emotion supporter, expert, facilitator, educator*) and evaluation metrics (EV); and 2) the design features (DF) (e.g., *On a scale from 1 to 5, please indicate the relevance of "adding real human support" to the design context*) and evaluation

	DF1 Adapt respond	DF2 Facilitate Q/As	DF3 Use of emojis	DF4 Steer chat	DF5 Add human profile	DF6 Embodied dialogue	DF7 Social dialogue	DF8 Small talks	DF9 Emotion smart	DF10 Align style	DF11 Social identity	DF12 Any other	DC2 (a) Info collect	DC2 (b) Emo support	DC2 (c) Local expert	DC2 (d) Service facilitate	DC2 (e) Safety educate	Count
EV1-Linguistic feature																		0
EV2-Time spent													0.267*			0.303*		2
EV3-Info effective								0.309*					0.270*	0.263*		0.483**	0.263*	5
EV4-Chat length																		0
EV5-Q/A quality		0.416**				0.296*	0.321*	0.271*		0.285*								5
EV6-Smoothness																		0
EV7-user entry time							0.304*						0.277*					2
EV8-error rate		0.266*															0.427**	2
EV9-dropout rate													0.361**				0.393**	2
EV10-system use																0.270*		1
EV11-Conv effective										-0.266*								1
EV12-Comm efficiency					0.300*												0.345**	2
EV13-Appropriateness																		0
EV14-Emotional smart	0.284*								0.398**	0.306*	0.489**			0.350**				5
EV15-Satisfaction				0.259*			0.293*											2
EV16-Privacy concern																	0.275*	1
EV17-Response quality																		0
EV18-Informativeness										0.260*								1
EV19-Ease of use																		0
EV20-Usefulness																		0
Count	1	2	0	1	1	1	3	2	1	4	1	2	3	1	0	5	3	NA

* p<0.05 ** p<0.01

Figure 3: Correlation of Design Features (DF) or Design Characteristics (DC) with Evaluation Metrics (EV)

metrics (EV). We discovered substantial relationships between some of the design choices (DC and DF) with the evaluation metrics.

More specifically, as shown in Fig 3, the DC with the highest correlation counts with EV is the *service facilitator* (DC2-d, count of intersections = 5); the DF with the highest correlation counts with EV is *aligned with the user's chat style* (DF10, count of intersections = 4).

Participants rated the three evaluation metrics – *information collect effectiveness* (EV3, i.e., the effectiveness of the chatbot in collecting safety related information from users' reports), *Q/A quality* (EV5, i.e., the perceived response quality of the chatbot reflected during the question/answer sessions), and *perceived emotional intelligence* (EV14, i.e., users' perception of the chatbot's emotional skills) as the evaluation metrics that have the highest correlations with DCs and DFs (count of intersections = 5 for all three metrics).

The perceived relevance of evaluation metrics may vary depending on context, and that some evaluation metrics may not be as important as previously suggested in prior literature. For example, the design feature DF10, which pertains to *aligning with an individual's chat style* (e.g., *language use, tone, accents, linguistic identity, codes used, timing and formality of language, etc.*), exhibited the highest correlation with the evaluation metrics in the context of safety reporting. Previous research has supported the efficacy of this design feature in the contexts of collaborative tasks [49], information providing [20], and language learning [1]. These prior studies evaluated chatbot systems on metrics such as trustworthiness, sense of control, conversation smoothness, and human-likeness. While our evaluation metrics did not include all of these factors, we did not detect similar evidence of *conversation smoothness* (EV6). These findings suggested that, by drawing insights from a user's perspective, the same design feature may have potential for new applications.

4.2 Preference for Empathetic Chatbot and its Relation with Design Features and Evaluation Metric

Prior result showed that the emotional intelligence (EV14, i.e., e.g., detecting, understanding, and responding to users' emotions appropriately, fosters users greater empathy with chatbot in conversations) has the highest correlation counts with DCs and DFs. Considering that EV14's main component is empathetic design [3, 39], which is closely related to the DC8 "*Would you like the chatbot to show empathy?*", we decided to further investigate how participants' preferences for DCs and DFs are in relations with their choices for EVs. This will aid in the comprehension of diverse user groups since, despite the fact that all users are participating in the design process, their varying criteria for defining the chatbot's design characteristics may result in varying evaluation metric preferences.

The result from Mann-Whitney test suggested that there is a significant difference in the perceived value of emotional competence (EV14) between participants who prefer a chatbot to display empathy (empathetic group) and those who do not (non-empathetic group). The Cliff's delta all showed medium to large effect size of the sample differences. Additionally, the empathetic group had a significantly higher weighting for certain design features (DF3, DF6, DF9, and DF10) compared to the non-empathetic group.

This implied that certain features may have an impact on users' *perceived emotional competence* (EV14) in the context of safety reporting. Previous literature has examined the effectiveness of similar features in other contexts, such as the *use of emojis* (EF3) to enhance perceptions of human-likeness [17, 52] and enjoyment [13], and the implementation of *emotional intelligence in chatbots* (DF9) to improve users' positive feelings [40] and productivity[17]. However, a review of existing literature on these features did not reveal any previous measurement of *perceived emotional competence*. Thus,

		Median (25 quantile, 75 quantile)		U	Mann-Whitney		Cliff's delta, effect size
		No(n=9)	Yes(n=49)		z	p	
Design Feature	DF3-Using emojis	1.000(1.0,1.0)	3.667(1.0,3.7)	76.5	-3.322	0.001**	-0.63, large
	DF6-Creating an empathetic embodied profile	2.333(1.0,3.7)	3.667(3.7,5.0)	125.5	-2.159	0.031*	-0.43, medium
	DF9-Demonstrating emotional intelligence	1.000(1.0,3.7)	5.000(3.7,5.0)	37	-4.377	0.000**	-0.83, large
	DF10-Being aligned to individual's chat style	3.667(1.7,3.7)	3.667(3.7,5.0)	128	-2.275	0.023*	-0.43, medium
Evaluation Metric	EV14- Perceived emotional competence	2.000(2.0,3.5)	4.000(4.0,5.0)	104	-2.663	0.008**	-0.53, large

*p<0.05 **p<0.01

Figure 4: Significant Variations in Design Feature (DF) and Evaluation Metric (EV) Preferences between Groups Determined by Attitude towards Empathetic Design (DC8: Yes vs No)

in the context of safety reporting, this EV might be a potential area for further exploration.

4.3 Early Chatbot Testing Increases Value for Evaluation Metric of *Ease of Use*

Workshop A and Workshop B differences in design characteristics (DC), design features (DF), and evaluation metrics (EV) were determined using chi-square and Mann-Whitney tests. The chi-square test revealed no significant differences in DC ($p>0.05$), while the Mann-Whitney test revealed non-significant differences in DF ($p>0.05$) but significant differences in EV ($p<0.05$). More specifically, the two group exhibits significance at the 0.05 level for *perceived ease of use* (EV19, $p = 0.049$, $U = 308$, $z = -1.969$), and there is a statistically significant difference between the two workshops: the median in the Workshop B/Fall 2022 ($n = 32$, median = 5.00) would be higher than the Workshop A/Spring 2022 ($n = 26$, median = 4.00) by a wide margin. The effect size ($d = 0.26$) calculate by the Cliff's delta package [21] further indicated the scores in Workshop A are slightly lower than the scores in Workshop B. This suggests that including a chatbot testing in the design process earlier on will encourage participants to place more value on the system's *ease of use* during the design activity.

5 DISCUSSION

Our case study shows that adding UX evaluation metrics (EV) as input material into the participatory design (generally applied at the effect phase of PD) enables us to understand users' values better and empower them in an early stage of the design process.

First, including UX evaluation metrics in the participatory design can help with rationalizing participants' design choices. On the one hand, as illustrated by the various correlation results in Section 4.1, there are clear correlations between the *input* and *output*. Thus, incorporating EVs into the participatory design input might aid practitioners in rationalizing participant design by revealing which evaluation metrics are significantly connected with participant preferences for which design options (DC and DF). This also indicated that the EV are effectively capturing design elements that are essential to users. On the other hand, by examining the most frequent intersection of EVs with DC and DF, designers can be better informed during the prioritization of design choices and evaluation metrics decision-making process.

Second, users can be empowered during the participatory design process because they can help extract key design values from the complexity of potential AI system effects when they define a system's evaluation metrics. We learn more about the various types of users by seeing how those who support empathic chatbot design and those who do not have distinct preferences for *perceived emotional competence* and certain DFs, such as emoji. This indicates a need for diversity in the customization of evaluation metrics so that the final design is inclusive and responsive to the needs of all users. In addition, non-AI systems utilize limited evaluation factors, such as usability and customer satisfaction [9, 23]. However, when developing AI systems and granting AI's capabilities, there are a great deal more evaluation metrics to consider [53]. If we can detect the design context and needs from the PD at an early stage, we can allow the user to determine the metrics they deem most essential. As in new domains, understanding where the (alternative) choices come from, how and by whom they are selected, and evaluated, is important [6].

Thirdly, when including UX evaluation metrics as a participatory design input, we found that the timing of testing an existing prototype affected the ease-of-use evaluation measure (EV). This indicates that familiarity with the design environment may play a moderating role in deciding the expected value of some participatory design effects. This factor can also be influenced by many other variables, which future research can validate, such as 1) whether the design itself is complex enough to elicit a difference in outcome based on the order of testing; 2) whether participants understand the design; and 3) whether the design prototype is detailed enough to reflect the changes that could occur based on the order of activities [26, 36].

Opportunities for future work include expanding the participant population to more diverse groups, beyond just HCI students, to validate and generalize the findings. In addition, there is a need to encourage collaboration between designers and end-users to define evaluation metrics that are inclusive and effective. Future research could explore how best to involve end-users in the *process* of defining metrics, as well as the impact of using the PD approaches on the design *effects*.

Lastly, to the best of our knowledge, we are the first to present an empirical example of the introduced usage of the *Program Theory Model* to explain PD [19], and our research added a new dimension to this mode by incorporating UX evaluation metrics into the PD

input phase. We look forward to seeing more studies to join the discussion of empowering participants in PD.

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